

QUIN HUTTON

Vehicle Dynamics Simulation | Damper & Ride Systems | Automotive Engineering Co-op

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VEHICLE DYNAMICS MODELS · MATLAB / SIMULINK

Shared parameter file; **Pacejka MF 6.1** tires from TTC data (Hoosier 43075); verified against closed-form limit cases.

2-DOF Bicycle (Milliken)	Steady-state yaw rate, sideslip and understeer gradient; aero load distribution swept across speed.
2-DOF Acceleration	Traction-limited launch with load transfer; drive-torque and grip-limited acceleration targets.
3-DOF Braking	Ideal brake-bias (I-curve) generation and per-axle lock-up margin.
2-DOF Quarter-Car	Ride and wheel-hop frequencies, damping ratios and tire normal-load variation under road input.
4-DOF Heave-Pitch (Half-Car)	Front/rear ride frequencies, pitch mode and heave-pitch coupling for spring-rate selection.
8-DOF Full-Car Ride	Per-corner loads with ARB rate, unsprung mass and chassis torsional compliance (split front/rear roll).
10-DOF Coupled Handling-Ride	Planar yaw/sideslip driven by instantaneous per-corner loads; steady-state and transient understeer.
1-DOF DSSV Damper	Force-velocity response of a DSSV spool-valve damper with asymmetric bump/rebound porting.

Supporting tools: QSS autocross lap sim (DXF tracks) calibrated to 2026 FSAE Michigan results; 15-parameter sensitivity study (PDY1/PDX1 dominant); Mahalanobis-distance tire matching for missing TTC data; FFT/frequency-sweep scripts.

WORK EXPERIENCE

Multimatic Ride Dynamics · Newmarket, ON May 2025 - September 2026
Process Engineer, Co-op

- **Line rebalance:** Designed a production-line rebalance with an estimated \$1.5M saving over the 8-year product life.
- **Vision inspection:** Raised Keyence EOL pass rate 11% by reducing measurement variance through parameter tuning and tooling changes, without relaxing quality thresholds.
- **Damper diagnostics:** Root-caused RPSD, BPSD and normal-position DSSV damper faults through systematic teardown; resolved valve-stack and internal component faults and documented corrective actions.
- **Damper modelling:** Built a 1-DOF DSSV model validated against production valve-code data, plus a MATLAB GUI port-window designer (open area vs. displacement, independent bump/rebound).
- **Automation:** Python tools for T1-2HD/LD dyno file management and TOX press CSV-to-Excel conversion.

Canadian Tire · Cocomile Service Centre · Lisgar Motors May 2021 - May 2025
Automotive Mechanic

- Diagnosed and repaired transmission, brake, suspension and electrical faults in high-volume service shops.

LEADERSHIP

MAC Formula Electric (FSAE) McMaster University Oct 2023 - Present
Vehicle Dynamics Lead (since June 2025)

- **Design process:** Own the 2027 VD model suite and design process, from tire data to spring, bias and aero targets.
- **Cross-team:** Set ride-rate, brake-bias and downforce targets for subteams via an 18-model roadmap with request intake.
- **Composites:** Built carbon/glass chassis parts: 75% lighter targeted part, part of a 35% chassis mass reduction.

TECHNICAL SKILLS

Software: MATLAB, Simulink, Pacejka MF 6.1, Python, Git, CATIA V5, SolidWorks; exposure to VI-grade, Concurrent Real-Time.
Engineering: DSSV diagnostics, valve modelling, DAQ, breakaway/residual torque, press-force, Keyence vision, GD&T, composites.

EDUCATION & AWARDS

McMaster University · Hamilton, ON Sept 2023 - Dec 2029 (expected)
B.Tech (Co-op), Automotive and Vehicle Engineering Technology - GPA 3.6/4.0

- Awards: Engineering Applicant Award (\$10,000); Most Inclusive Design, McMaster Welcome Week; Entrance Award (\$2,000).
- Motorsport Engineer course: Vehicle Dynamics, Modelling & Simulation (2025).